Corey Birnie

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WORK HISTORY

Worldover

Advanced Software Engineer **January 2026 – Present**

- Headed a team of engineers through the architectural transformation of a legacy platform into an "AI-first" app-factory, enabling the automated generation of bespoke client web apps and overseeing the full lifecycle from research to execution.
- Directed a team of engineers to architect and launch 'QuickReview,' a flagship greenfield AI feature; maintained full project ownership to more than 100x customer efficiency on 'End-product review'.
- Built a document agent enabling customers to create specialised, reusable document generators, turning bespoke one-off document work into a repeatable self-serve capability.
- Built a request agent that uses agent mail to autonomously chase and retrieve documents from suppliers over email, then extract structured data from them, removing a manual back-and-forth from the customer workflow.

Full-Stack Software Engineer **August 2024 – January 2026**

- Implemented an MCP server on the platform API during personal time, creating a foundation for integrating AI agent-based features across the platform
- Mentored new engineers by managing their task progression, accelerating their integration into the development cycle.
- Engineered a substantial portion of the platform's core architecture and feature set as an early-stage hire, driving the product's evolution from its foundational stages to a production-ready enterprise platform.

Citigroup – *Software Engineer Intern* **June 2023 – August 2023**

- Worked with Python and Tableau to analyse feature bias in predictive Machine Learning models used to determine loan acceptance
- Developed scripts using PL/SQL to onboard a new compliance feature, and Java to improve automation on updated systems
- Received mentorship and training in quantitative analysis and technical algorithms

Durham University – *Demonstrator/Teaching Assistant* **September 2022 – June 2024**

- Led and taught groups of 30 first- and second-year students in a mixture of practical programming and AI modules
- Learned how to communicate my knowledge to less experienced students in an easy-to-understand manner to help them learn

MBDA – *Software Engineer Intern* **June 2021 – August 2021**

- Developed embedded real-time systems using C/C++ with high security and reliability standards, gaining experience in low-level development and comprehensive testing methods
- Implemented static analysis tooling, configured build systems while writing unit tests in C++ and Python

EDUCATION

Durham University – *MEng Computer Science* **June 2024**

1st class honours | Durham Inspired Scholarship recipient

Activities: DU Computing Society, Durhack, Grey College Boat Club, DU Brazilian Jiu Jitsu, DU Lifting Club

New College Doncaster – *A Levels* **June 2020**

A-Levels: Mathematics (A*), Computer Science (A*), Physics (A), AS Chemistry (B)

PROJECTS

Ticketing System on the IC Blockchain

- Co-founded a Web3 startup with two engineering students, securing grant funding from the DFINITY Foundation to architect a decentralised event ticketing platform on the Internet Computer blockchain.
- Developed and published open-source Motoko/Rust algorithms, including dynamic on-chain NFT QR code generation to eliminate secondary market ticket fraud.

Master's Dissertation: Detecting CEO Behaviour from Video Interviews via ML

- Employed advanced computer vision techniques in tandem with machine learning models for precise facial expression

detection

- Integrated facial expression data with additional visual indicators (head position, eye movement, etc.) and NLP data into a tailored machine learning model to accurately forecast future stock prices
- Deployed a multi-modal approach to improve the accuracy of stock forecasting models

Image Generative AI Model: Novel GAN for Image Synthesis

- Constructed a novel Generative Adversarial Network (GAN) from scratch, capable of generating coherent images from random noise inputs
- Executed the project using PyTorch, demonstrating proficiency in harnessing advanced machine learning libraries
- Integrated concepts from diverse research studies to optimise the model, achieving state-of-the-art performance in image generation

SKILLS

Languages: TypeScript, Python, SQL, Java, C++, Motoko, Haskell

Technologies: React, Next.js, tRPC, Zod, Postgres, Tailwind, Framer Motion, CSS, React Query, Git, Linux, PyTorch, ThreeJS, MCP

Infrastructure: AWS(EC2, Lambda, S3, IAM, Amplify, Load Balancer, ...), Supabase, Vercel, Firecracker VMs, Authentication Integration